

Semester IV
Credit and Examination Scheme

Credit Scheme

Course Code	Course Name	Teaching Scheme (Hrs.) Per Week TH – P – TUT	Total (Hrs.) Per week	Credits Assigned TH – P – TUT	Total Credits	Course Category
116U01C401	Probability, Statistics and Optimization Techniques ^{\$}	3 – 0 – 1	4	3 – 0 – 1	4	BS
116U01C402	Analysis of Algorithms	3 – 0 – 0	3	3 – 0 – 0	3	PC
116U01C403	Relational Database Management Systems	3 – 0 – 0	3	3 – 0 – 0	3	PC
116U01C404	Theory of Automata with Compiler Design	3 – 0 – 1	4	3 – 0 – 1	4	PC
116U01L401	Web Programming Laboratory	0 – 4 – 0	4	0 – 2 – 0	2	PC
116U01L402	Analysis of Algorithms Laboratory	0 – 2 – 0	2	0 – 1 – 0	1	PC
116U01L403	Relational Database Management Systems Laboratory	0 – 2 – 0	2	0 – 1 – 0	1	PC
116U01P401	Mini Project	1 – 2 – 0	3	0 – 3 – 0	3	PR
	Total	13 – 10 – 2	25	12 – 7 – 2	21	
116U01A401	Audit Course	2 – 0 – 0	2	--	--	AC

\$- Common with IT Branch

Examination Scheme

Course Code	Course Name	Examination Scheme							
		Marks							
		CA		ESE	TW	O%	P	P&O [#]	Total
		ISE	IA						
116U01C401	Probability, Statistics and Optimization Techniques ^{\$}	30	20	50	25	-	-	-	125
116U01C402	Analysis of Algorithms	30	20	50	-	-	-	-	100
116U01C403	Relational Database Management Systems	30	20	50	-	-	-	-	100
116U01C404	Theory of Automata with Compiler Design	30	20	50	25	-	-	-	125
116U01L401	Web Programming Laboratory	-	-	-	50	-	-	50	100
116U01L402	Analysis of Algorithms Laboratory	-	-	-	25	-	-	25	50
116U01L403	Relational Database Management Systems Laboratory	-	-	-	25	-	-	25	50
116U01P401	Mini Project	-	-	-	50	-	-	50 [^]	100
	Total	120	80	200	200		-	150	750
116U01A401	Audit Course	-	-	-	-	-	-	-	-

% Oral examination based on entire theory syllabus, # based on practical & the corresponding theory Syllabus \$- Common with IT Branch ^Demo based on mini project and viva based on implementation

Course Code	Course Title							
116U01C401	Probability, Statistics and Optimization Techniques							
	TH			P	TUT			Total
Teaching Scheme(Hrs.)	03			--	01*			04
Credits Assigned	03			--	01			04
Examination Scheme	Marks							
	CA		ESE	TW	O	P	P&O	Total
	ISE	IA						
	30	20	50	25	--	--	--	125

* Batch wise Tutorial

Course prerequisites:

- Basics of Statistics and Probability
- Introductory Linear programming problems

Course Objectives

This course Exposes students to the concepts of Correlation, Regression for given bivariate data. Students are made familiar with different discrete and continuous probability distributions. The course acquaints students with concepts of Large sample test, Small sample test and Chi – Square test. The course familiarizes students with different methods of solving Linear and Nonlinear Programming problems. Some basic queuing theory models are also discussed in the course. Using these methods it will be possible to analyze and interpret a given real life situation and think of possible solutions.

Course Outcomes

At the end of successful completion of the course the student will be able to

- CO1. Apply concepts of correlation, regression for given bivariate data.
- CO2. Apply concepts of Binomial, Poisson, Exponential and Normal distribution to solve Engineering problems.
- CO3. Apply Large sample test and small sample test to analyze collected data.
- CO4. Apply concepts of Linear and Nonlinear programming methods to solve problems.
- CO5. Apply the methods of single server limited queue and single server unlimited queue models to analyze and interpret the data.

Module No.	Unit No.	Details	Hrs.	CO
1	Probability and Probability Distribution		12	CO 1
	1.1	Conditional Probability, Bayes' theorem, Joint Probability		
	1.2	Discrete and Continuous Probability Distribution		
	1.3	Binomial Distribution, Poisson Distribution		
	1.4	Uniform Distribution, Normal Distribution, Exponential Distribution		
2	Correlation and Regression		06	CO 2
	2.1	Correlation, Co-variance, Karl Pearson Coefficient of Correlation & Spearman's Rank Correlation Coefficient.		
	2.2	Regression Coefficients, lines of regression & logistic regression		
3	Sampling Theory		07	CO 3
	3.1	Sampling distribution. Test of Hypothesis. Level of significance, critical region. One tailed and two tailed tests. Interval Estimation of population parameters. Large and small samples.		
	3.2	Difference between sample mean and population means for large samples, Test for significance of the difference between the means of two large samples.		
	3.3	Student's t-distribution: Test for significance of the difference between sample mean and population means, Test for significance of the difference between the means of two Samples, paired t-test.		
	3.4	Chi-square distribution as a Test of Independence, Test of the Goodness of fit and Yate's correction.		
	3.5	Fisher's z-test		
4	Optimization Techniques		13	CO 4
	4.1	Types of solution, Standard and Canonical form of LPP, Basic and feasible solutions, simplex method.		
	4.2	Artificial variables, Big -M method (method of penalty).		
	4.3	Duality and Dual Simplex method		

	4.4	Unconstrained optimization, problems of two or three variables with one equality constraint using Lagrange's Multiplier method.		
	4.5	Problems of two or three variables with one inequality constraint using Kuhn-Tucker conditions		
5	Queuing Theory		07	CO5
	5.1	Introduction, Features of Queuing, solution of Queuing models. M/M/1(Singal Server, Unlimited Queue Model)		
	5.2	M/M/1 Singal Server, limited Queue Model		
Total			45	

Recommended Books:

Sr. No.	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
1.	B. S. Grewal	<i>Higher Engineering Mathematics</i>	Khanna Publications, India	43 rd Edition 2014
2.	Erwin Kreyszig	<i>Advanced Engineering Mathematics</i>	Wiley Eastern Limited, India	10 th Edition 2015
3.	J. K. Sharma	<i>Operation research: Theory and Applications</i>	Laxmi Publications, India	6 th Edition 2017
4.	S.C.Gupta and V.K.Kapoor	<i>Fundamentals of Mathematical Statistics</i>	Sultan Chand & Sons	11 th Edition 2009
5.	Ronald E. Walipole, Raymond H.Myers	<i>Probabilities & Statistics for Engineers & Scientists</i>	Pearson Education	9 th Edition 2010
6.	P. N. Wartikar and J. N. Wartikar	<i>A textbook of Applied Mathematics Vol I & II</i>	Pune Vidyarthigruha, India	6 th Edition 2012

Term-Work will consist of Tutorials covering the entire syllabus. Students will be graded based on continuous assessment of their term work. **At least 2 tutorials will be conducted with the help of Mathematical and Statistical software in the Laboratory.**

Course Code	Course Title							
116U01C402	Analysis of Algorithms							
	TH			P		TUT		Total
Teaching Scheme (Hrs./Week)	03			02		--		05
Credits Assigned	03			01		--		04
Examination Scheme	Marks							
	CA		ESE	TW	O	P	P&O	Total
	ISE	IA						
	30	20	50	25	--	--	25	150

Course prerequisites:

Data structure and Discrete Structures.

Course Objectives:

The objective of the course is to teach various techniques for effective problem solving in computing. The different algorithm paradigms for problem solving will be used to illustrate efficient methods to solve problems. The analysis of the algorithm will be demonstrated to show the efficiency of the algorithm. The complexity theory of the problems is introduced to students for further analysis of algorithms.

Course Outcomes

At the end of successful completion of the course the student will be able to

- CO1:** Analyze the asymptotic running time and space complexity of algorithms.
- CO2:** Describe various algorithm design strategies to solve different problems and analyze complexity.
- CO3:** Develop string matching techniques
- CO4:** Describe the classes P, NP, and NP-Complete

Module No.	Details	Hrs.	CO
1	Introduction to analysis of algorithm Performance analysis, space and time complexity, Growth of function-Big-Oh; Omega; Theta Notation, Analysis of insertion sort, Introduction to randomized algorithm, Solving Recurrence Problems by Substitution Method, Recursion Tree Method, Masters Method	06	CO1
2	Algorithm Design Techniques		
	Divide and Conquer Technique General method, Finding minimum and maximum algorithm and analysis, Analysis of Merge sort, Analysis of Quick sort, Fast Fourier Transform, Pattern matching using divide and conquer	06	CO2
	Greedy Technique General method, Knapsack problem, Minimum cost spanning trees-Kruskal's and Prims algorithm, Single source shortest path	06	CO2
	Dynamic Programming Technique General method, Multistage graphs, 0/1 knapsack, Travelling salesman problem, Single source shortest path, All pairs shortest path, Matrix chain multiplication	07	CO2
	Backtracking Technique General method, Sum of subsets, 8 queens problem, Graph coloring	04	CO2
	Branch and Bound General method, 0/1 Knapsack, 15 Puzzle Problem	04	CO2
3	String Matching Algorithms The naïve string-matching Algorithms, String matching with finite automata, The Knuth-Morris-Pratt algorithm, Longest common subsequence	06	CO3
4	Non-deterministic Polynomial Algorithms Polynomial time, Polynomial time verification, NP Completeness and reducibility, NP Completeness proof: Vertex Cover Problem, Clique Problem	06	CO4
	#Self-Learning Topic- Rod cutting algorithm, randomization algorithms		
TOTAL		45	

Students should prepare all Self Learning topics on their own. Self-learning topics will enable students to gain extended knowledge of the topic. Assessment of these topics may be included in IA.

Recommended Books:

Sr. No.	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
1	T. H. Coreman, C.E. Leiserson, R.L. Rivest, and C. Stein	<i>“Introduction to Algorithms”</i> ,	PHI Publication	2 nd Edition, 2005
2	Ellis Horowitz, Sartaj Sahni, S. Rajsekaran,	<i>“Fundamentals of Computer Algorithms”</i>	University Press	2 nd Edition, 2008
3	Alfred V. Aho, John E. Hopcroft, Jeffrey D. Ullman	<i>“Data Structures and Algorithm”</i>	Pearson education	4 th Impression 2009
4	Michael Gooddrich & Roberto Tamassia	<i>“Algorithm Design Foundation, Analysis and Internet Examples”</i>	Wiley Student Edition	2nd Edition.

Course Code	Course Title							
116U01C403	Relational Database Management Systems							
	TH			P	TUT			Total
Teaching Scheme (Hrs./Week)	03			02	--			05
Credits Assigned	03			01	--			04
Examination Scheme	Marks							
	CA		ESE	TW	O	P	P&O	Total
	ISE	IA						
	30	20	50	25	-	-	25	150

Course prerequisites:

Data Structure and programming knowledge

Course Objectives:

The objective of the course is to design and program database systems. It covers ER (Entity-Relationship) approach to data modeling, the relational model of Database systems (DBMS) and efficient database design using normalization. It covers Relational Algebra and use of Query Languages such as SQL. This course also introduces Transaction Management, Concurrency Control and Recovery Techniques. The course achieves balance between firm theoretical foundation to designing moderate size databases and creating, querying and implementing realistic databases.

Course Outcomes:

At the end of successful completion of the course the student will be able to

CO1: Understand the features of Relational database management systems.

CO2: Develop relational database design using the designed Entity-Relationship model.

CO3: Use SQL for Relational database creation, maintenance and query processing

CO4: Understand and analyze indexing, hashing, Query processing, query optimization, and Normalization of relational database.

CO5: Apply the transaction, concurrency and recovery techniques

Module No.	Unit No.	Details of Topic	Hrs.	CO
1	Introduction		05	CO1
	1.1	Introduction, Characteristics of databases, Comparison of File system and Database approach, Users of Database system, Concerns when using an enterprise database		
	1.2	Data Independence, DBMS system architecture, Database Administrator		
2	Data Modeling: Enhanced-Entity-Relationship Model and Relational Data Model		10	CO2
	2.1	Introduction, Benefits of Data Modeling, Types of Models, Phases of Database Modeling, The Entity-Relationship (ER) Model		
	2.2	Enhanced -Entity-Relationship (EER)- Model Generalization, Specialization and Aggregation		
	2.3	Relational Model: Introduction, Data Manipulation, Data Integrity, Advantages of the Relational Model		
	2.4	Mapping EER Model to Relational Model		
3.	Relational Algebra and Structured Query Language (SQL),		08	CO2
	3.1	Relational Algebra, Relational Algebra Queries		
	3.2	Overview of SQL, Data Definition Commands, Domain Constraints, Referential integrity		
	3.3	Set operations, aggregate function, null values, Data Manipulation Commands		
	3.4	Data Control commands, Views in SQL, Nested and complex queries, Assertions, Trigger, Security and authorization in SQL		
4	Query Processing and optimization		08	CO4
	4.1	Indexing: Basic concepts, ordered indices: dense and sparse, multilevel indices, secondary indices		
	4.2	Hashing: Static hashing, dynamic hashing, comparison of ordered indexing and hashing		
	4.3	Query processing: Steps involved in query processing, measures of query cost, algorithms for SELECT and PROJECT operations.		
	4.4	Optimization: Overview, Transformation of relational expressions, Estimating statistics, Choice of evaluation plan		

5	Relational–Database Design		07	CO4
	5.1	First Normal Form, Pitfalls in Relational-Database designs		
	5.2	Function Dependencies, Armstrong Axioms		
	5.3	2nd, 3rd, BCNF and 4th normal form		
	5.4	Decomposition, desirable properties of decomposition		
	5.5	Overall database design process		
6	Transaction Management, Concurrency control and Recovery protocols		07	CO5
	6.1	Transaction concept, Transaction states, ACID properties		
	6.2	Characterizing schedule based on recoverability and serializability		
	6.3	Concurrency Control: Two-Phase Lock-based, Timestamp-based, Multi-version Concurrency Control, Validation-based protocols, Deadlock Handling-Wait for graph		
	6.4	Recovery System: Recovery concept, Log based recovery, Shadow paging		
Total			45	

Recommended Books:

Sr. No.	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
1	Elmasri and Navathe	<i>“Fundamentals of Database Systems”</i>	Pearson Education	6 th Edition
2	Korth, Silberchatz, Sudarshan	<i>“Database System Concepts”</i>	McGraw Hill	6 th Edition
3	Raghu Ramakrishnan, Johannes Gerhke	<i>“Database Management Systems”</i>	McGraw Hill	6 th Edition
4	G. K. Gupta	<i>“Database Management Systems”</i>	McGraw Hill.	6 th Edition

Course Code	Course Title							
116U01C404	Theory of Automata with Compiler Design							
	TH		P		TUT		Total	
Teaching Scheme (Hrs./Week)	03		--		01		04	
Credits Assigned	03		--		01		04	
Examination Scheme	Marks							
	CA		ESE	TW	O	P	P&O	Total
	ISE	IA						
	30	20	50	25	--	--	--	125

Course prerequisites:

Students should be familiar with concepts of discrete structures.

Course Objectives

Aims to build concepts regarding the fundamental principles of Grammars, Automata Theory, Turing Machines, PushDown Automata, Un-decidability and Intractable Problems. It aims to understand the design of computing machines that can perform complex computation.

Course Outcomes

At the end of successful completion of the course the student will be able to

- CO1:** Describe regular languages using Regular Expressions, Finite Automata, Nondeterministic Finite Automata, Mealy Machines, Moore Machines and its applications
- CO2:** Write, simplify and normalize context free grammars and describe context free languages using context free grammar and push down automata
- CO3:** Design Turing Machines for various problems and its applications
- CO4:** Understand the concept of Un-decidability and Recursively Enumerable Languages

Module No.	Unit No.	Details	Hrs.	CO
1	Finite Automata		08	CO1
	1.1	Introduction: Alphabets, String, Language, Basic Operations on language, Concatenation, Kleene Star Introduction to different phases of compiler.		
	1.2	Finite Automata (FA) -its behavior; DFA -Formal definition, state transition diagram, transition table, Language of a DFA. NFA -Formal definition, state transition diagram, transition table Language of an NFA. FA with epsilon-transitions, Eliminating epsilon-transitions, Equivalence of DFAs and NFAs, Conversion from NFA to DFA. Moore machine and Mealy Machine- Formal definition, state transition diagram, transition table, Conversion from Mealy to Moore machine and Moore to Mealy machine. Application of Finite Automata for Lexical Analysis and Lex tools		
2	Regular Languages		09	CO1
	2.1	Chomsky hierarchy, Regular sets, Regular Expression, Some closure properties of Regular languages -Closure under Boolean operations, reversal, homomorphism, inverse homomorphism, etc..		
	2.2	FA and Regular Expressions, equivalence between FA and regular expressions		
	2.3	Pumping lemma for Regular languages, Equivalence and minimization of Finite Automata, Myhill-Nerode Theorem		
	2.4	Application of finite automata and regular expression in lexical analysis		
3	Context Free Grammars		07	CO2
	3.1	Context-free Grammars (CFGs) -Formal definition, sentential forms, leftmost and rightmost derivations, the language of a CFG. Derivation tree or Parse tree-Definition, Simplification of CFGs -Removing useless symbols, epsilon-Productions, and unit productions		
	3.2	Relationship between parse trees and derivations. Parsing and ambiguity, Application of CFGs, Ambiguity in grammars and Languages.		
	3.3	Normal forms -CNF and GNF. Proving that some languages are not context free -Pumping lemma for CFLs, applications. Some closure properties of CFLs -Closure under union, concatenation, Kleene closure, substitution, Inverse homomorphism, reversal, intersection with regular set, etc. Some more decision properties of CFLs		

4	Push Down Automata		07	CO2
	4.1	Pushdown Automata (PDA) -Formal definition, behavior and graphical notation, Instantaneous descriptions (Ids),		
	4.2	The language of PDA (acceptance by final state and empty stack). Equivalence of acceptance by final state and empty stack, Equivalence of PDAs and CFGs,		
	4.3	Conversion: CFG to PDA, PDA to CFG.		
	4.4	DPDAs -Definition, DPDAs, Multi-stack DPDAs & NPDAs and CFLs, Languages of DPDAs, NPDAs		
5	Turing Machine		10	CO3
	5.1	Turing Machines TM -Formal definition and behavior, Transition diagrams, Language of a TM, TM as accepters deciders and generators. TM as a computer of integer functions, Design of TMs, Programming techniques for TMs - Storage in state, multiple tracks, subroutines, etc.		
	5.2	Universal TMs, Variants of TMs –Multi-tape TMs, Nondeterministic TMs. TMs with semi-infinite tapes, Multi-stack machines, Simulating TM by computer, Simulating a Computer by a TM		
	5.3	Equivalence of the various variants with the basic model.		
	5.4	Introduction to parsers, types of parser		
	5.5	Application of CFG, PDA, TM in parsing		
6	Un-decidability and Recursively Enumerable Languages:		04	CO4
	6.1	Recursive and Recursively Enumerable Languages. Properties of Recursive and Recursively Enumerable Languages.		
	6.2	Decidability and Undecidability, Halting Problem, Rice’s Theorem, Greibach’s Theorem, Post Correspondence Problem, Context Sensitivity and Linear Bound Automata.		
TOTAL			45	

Recommended Books:

Sr. No.	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
1.	John E. Hopcroft, Rajeev Motwani, Jeffrey D. Ullman	“Introduction to Automata Theory, Languages and Computation”	Pearson Education	Third Edition, 2006
2.	J.C.Martin,	“Introduction to languages and the Theory of Computation”	TMH	Fourth Edition, 2010
3.	Michael Sipser	“Theory of Computation”	Cengage Learning	Third Edition, 2012
4.	O.G.Kakde	“Theory of Computation”	Laxmi Publications	First Edition, 2007

Term-Work will consist of Tutorials covering the entire syllabus. Students will be graded based on Continuous Assessment of their term work.

Course Code	Course Title							
116U01L401	Web Programming Laboratory							
	TH		P	TUT			Total	
Teaching Scheme(Hrs.)	--		04	--			04	
Credits Assigned	--		02	--			02	
Examination Scheme	Marks							
	CA		ESE	TW	O	P	P&O	Total
	ISE	IA						
	--	--	--	50	--	--	50	100

Course prerequisites:

Basic Programming skills.

Course Objectives

Objective of this course is to provide students an overview of the concepts required for development of application based on Web Technologies.

Course Outcomes

At the end of successful completion of the course the student will be able to

- CO1. Design dynamic web pages using various HTML tags.
- CO2. Use CSS to prepare the layout of web pages.
- CO3. Apply JavaScript for validation in client side programming.
- CO4. Integrate server side pages using php.
- CO5. Apply database operations by integrating SQL queries and session variables.

Module No.	Unit No.	Details	Hrs.	CO
1	HTML,DHTML			
	1.1	Designing of effective web site, Introduction to different Web Technologies. HTML Tag Reference, Global Attributes, Event Handlers, Document Structure Tags, Formatting Tags, Text Level formatting, Block Level formatting, List Tags, Hyperlink tags, Image and Image maps,	4	1
	1.2	Table tags, Form Tags, Frame Tags, Executable content tags.		
2	CSS and Bootstrap		4	2
	2.1	What are style sheets?, Why are style sheets valuable? Different approaches to style sheets, Using Multiple approaches, Linking to style information in separate file, Setting up style information, Using the <LINK> tag, embedded style information, Using <STYLE> tag, Inline style formation		
	2.2	Introduction to Bootstrap, Bootstrap grids, layouts, bootstrap components like iconography, dropdowns, input groups, navigation, alerts. and plugins		
3	JavaScript		6	3
	3.1	Introduction to JavaScript, Data Types, Operators, Control Flow, Arrays, and Functions		
	3.2	Making Decisions / Repeating Code; Debugging and Error Handling; Working with DOM and DHTML		
	3.3	Enhancing and Validating Forms		
4	PHP Programming		7	4
	4.2	PHP : Why PHP and MySQL?, Server-side web scripting, Installing PHP, Adding PHP to HTML, Syntax and Variables, Passing information between pages, Strings, Arrays and Array Functions, Numbers, Handling basic PHP errors / problems.		
5	PHP and MySQL		9	5
	5.1	PHP/MySQL Functions, Displaying queries in tables, Building Forms from queries, String and Regular Expressions, Sessions, Cookies, Integration of complete web application and deployment.		
	5.2	#self learning topic: Study of Laravel framework		
Total			30	

Recommended Books:

Sr. No.	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
1.	Thomas Powell	Web Design The complete Reference	Tata McGrawHill	5 th edition 2010
2.	Thomas Powell	HTML and XHTML The complete Reference	Tata McGrawHill	5 th edition 2010
3.	Thomas Powell and Fritz Schneider	JavaScript 2.0 : The Complete Reference,	Tata McGrawHill	3rd Edition,2013
4.	Steven Holzner	PHP : The Complete Reference	Tata McGrawHill	2 nd edition 2008

Term-Work consists of programming assignments covering entire syllabus. Students will be graded based on continuous assessment of their term work.

Course Code	Course Title							
116U01P401	Mini Project							
	TH		P		TUT		Total	
Teaching Scheme (Hrs.)	01		02		--		03	
Credits Assigned	--		03		--		03	
Examination Scheme	Marks							
	CA		ESE	TW	O	P	P&O	Total
	ISE	IA						
	-	-	-	50	-	-	50	100

Course Objectives:

The objective of Mini Project is to address a real-world problem, find, implement and demonstrate the solution for the same through the courses learned in earlier semesters. Identify various hardware and software requirements for problem solution. It will also inculcate qualities such as meeting deadlines, making and following work plan. The Mini Project may be beyond the scope of courses learnt and interdisciplinary in nature.

Course Outcomes:

After successful completion of the course student should be able to

- CO1. Understand the requirements for problem definition and scope.
- CO2. Identify the various hardware and software, usage of Tools needed to meet the desired specification.
- CO3. Describe the design in the form of algorithm/flowchart/block diagram.
- CO4. Implement and test the design as per mentioned specifications.
- CO5. Prepare a technical report based on Mini project.
- CO6. Present technical seminar based on the Mini project work carried out.

Course delivery guidelines related to Software Engineering principles:

In Theory class teacher is expected to discuss and demonstrate the SDLC process based on waterfall Model.

Module No.	Unit No.	Details	Hrs.	CO
1	Understand the problem definition (Analysis phase)		03	CO1
	1.1	Identify users (different users/stakeholders)		
	1.2	Identify role of every user. who and for what ? (view user, transaction user, admin user etc.)		
	1.3	Is there any parallel system? (survey concept.)		
	1.4	Demonstrate the systems known to students like LMS, MS-Teams, Somaiya web site, with which students are familiar, or the demos prepared/compiled by faculty		
	1.5	Identify need of the project		
	1.6	Prepare problem definition, including background, and scope of the work		
2	Identification of data and hardware software Requirement (Analysis phase)		02	CO2
	2.1	Identify data		
	2.2	Identify software , and hardware needed		
	2.3	Prepare E-R schema or equivalent description of data		
	2.4	Identify usage of Tools, library functions, APIs/packages, applicable to solve the problem		
3	Identify Design description (Design phase)		03	CO3
	3.1	Identify Design description		
	3.2	Represent the definition in the form of block diagram, flowchart, use case diagram		
	3.3	Identify flow of information		
	3.4	Represent information flow in the graphical diagram (web graph, flowchart etc.)		
	3.5	Design user interfaces as per role defined in 1.3 (UI with HTML or equivalent Tool		
	3.6	Describe data in details (e g E-R to Relational Schema)		
	3.7	Define functions as per mentioned scope		
4	Implementation phase (Coding phase)		03	CO4
	4.1	Coding of different modules as defined in the scope		
	4.2	Integration of modules (as per 3.3)		
	4.3	Identify test cases related to different modules.		
	4.4	Prepare test cases		
	4.5	Prepare responsibility chart (Test case writer, Tester) in the form of table		

5	Testing phase		01	CO4
	5.1	Test the modules as per written test case		
	5.2	Complete the test case chart (4.5 - responsibility chart)		
6	Deployment Phase		01	CO4
	6.1	Describe need of deployment		
	6.2	Study of GitHub		
	6.3	Upload the project on GitHub		
7	Preparation of Report		02	CO5
	7.1	Understand the different sections of Report template.		
	7.2	Write report specifying points 1 to 6 in corresponding section		
	7.3	Describe contents of Presentation		
8	Preparation of Presentation for final demonstration as per report sequence			CO5
Total			15	

Term Work and Practical / Oral:

The mini project is a group project. Interdisciplinary projects are also permitted. Each project will be assigned to one faculty member as a supervisor.

There will be continuous assessment and progress report of the project that needs to be maintained by student(s). The final oral will be a presentation based on a demonstration of the project in front of a committee of examiners.

Course Code	Course Title							
116U01L402	Analysis of Algorithms Laboratory							
	TH		P	TUT			Total	
Teaching Scheme(Hrs.)	--		02	--			02	
Credits Assigned	--		01	--			01	
Examination Scheme	Marks							
	CA		ESE	TW	O	P	P&O	Total
	ISE	IA						
	--	--	--	25	--	--	25	50

Term work will consist of experiments based on the syllabus of “Analysis of Algorithm (116U01C402)”. Students will be graded based on the continuous assessment of their term work. Practical and Oral examination will based on the syllabus of “Analysis of Algorithms” and laboratory experiments.

Course Code	Course Title							
116U01L403	Relational Database Management Systems Laboratory							
	TH			P		TUT		Total
Teaching Scheme(Hrs.)	--			02		--		02
Credits Assigned	--			01		--		01
Examination Scheme	Marks							
	CA		ESE	TW	O	P	P&O	Total
	ISE	IA						
	--	--	--	25	--	--	25	50

Term work will consist of experiments based on syllabus of ‘Relational Database Management System’ (116U01C403). Students will be graded based on continuous assessment of their term work. Oral and Practical examination will be based on syllabus of ‘Relational Database Management System’ and laboratory experiments.